BONACA · VILLA WEBSITE & CRM

Environment **Setup Guide**

Every credential still missing from `.env`, where to click to get it, and exactly what to paste back. Written against the code as it stands today.

Project	bonaca · Next.js 16.3.4 · Firebase Admin + Cloudflare R2
Firebase project	`bonaca-5689e`
Values still pending	**7** — 2 Firebase, 5 Cloudflare R2
Removed as unused	7 `NEXT_PUBLIC_FIREBASE_*` keys
Time needed	~10 min Firebase · ~15 min Cloudflare R2
Generated	7 September 2026

0 Where things stand

I read every `process.env` reference in the codebase and compared it against your `.env`. This is the complete picture — there are no other variables the app looks for.

Status of all 15 variables

VARIABLE	STATUS	WHERE THE VALUE COMES FROM
`ADMIN_USERNAME`	SET	Your choice — currently `admin`
`ADMIN_PASSWORD`	WEAK	Your choice — see Part 4, currently 12345678
`ADMIN_SESSION_SECRET`	SET	Random 32-byte hex, already generated
`FIREBASE_PROJECT_ID`	SET	`bonaca-5689e`
`FIREBASE_CLIENT_EMAIL`	PENDING	**Part 1** — service-account JSON
`FIREBASE_PRIVATE_KEY`	PENDING	**Part 1** — same JSON file
`FIREBASE_CONTENT_COLLECTION`	SET	Your choice — `bonaca-cms`
`FIREBASE_CONTENT_DOCUMENT`	SET	Your choice — `site-content`
`R2_ACCOUNT_ID`	PENDING	**Part 2.3** — Cloudflare R2 sidebar
`R2_ACCESS_KEY_ID`	PENDING	**Part 2.4** — R2 API token
`R2_SECRET_ACCESS_KEY`	PENDING	**Part 2.4** — same token, shown once
`R2_BUCKET`	PENDING	**Part 2.2** — bucket name you pick
`R2_PUBLIC_BASE_URL`	PENDING	**Part 2.5** — public bucket URL
`R2_PREFIX`	SET	Folder inside bucket — `bonaca`
`NEXT_PUBLIC_SITE_URL`	LATER	localhost now; real domain at deploy time

What I removed from .env, and why

These seven keys were in your file but **not referenced anywhere in the codebase**:

```
# deleted — nothing reads these
NEXT_PUBLIC_FIREBASE_API_KEY
NEXT_PUBLIC_FIREBASE_AUTH_DOMAIN
NEXT_PUBLIC_FIREBASE_PROJECT_ID
NEXT_PUBLIC_FIREBASE_STORAGE_BUCKET
NEXT_PUBLIC_FIREBASE_MESSAGING_SENDER_ID
NEXT_PUBLIC_FIREBASE_APP_ID
NEXT_PUBLIC_FIREBASE_MEASUREMENT_ID
```

They belong to the Firebase *web* SDK. This project only uses `firebase-admin` (server-side), and the `firebase` client package is not even installed. Keeping them meant seven public values shipped to every visitor's browser for no reason. A timestamped backup of the original file sits beside `.env` if you ever want them back.

> **Nothing breaks by deleting them**
> The CRM reaches Firestore through the service account in Part 1. If you later add Google Analytics or client-side Firebase Auth, you would re-add only the specific keys that feature needs.

How the app behaves right now, with these blank

The code degrades gracefully rather than crashing — that is deliberate:

- **Firebase blank** → content saves to `.data/content.json` on your laptop. Edits in `/admin` work, but never leave your machine and are lost on a fresh deploy.
- **R2 blank** → uploaded photos land in `public/uploads/`. Fine locally; on Vercel these vanish on every deploy because the filesystem is read-only and ephemeral.

So both parts below are genuinely required before the site goes live — but neither blocks you from building today.

1 Firebase — 2 values

One download gives you both `FIREBASE_CLIENT_EMAIL` and `FIREBASE_PRIVATE_KEY`. They come out of the same JSON file, so this is really one job, not two.

1.1 Confirm the Firestore database exists

The service account is useless without a database to talk to. Open the link, and if you see a **“Create database”** button, click it — it has not been made yet.

> OPEN FIRESTORE
> https://console.firebase.google.com/project/bonaca-5689e/firestore

1. Click **Create database** (skip this if you already see a data table).
2. Choose **Start in production mode**. The Admin SDK bypasses security rules entirely, so locked-down rules will not block your CRM — they only block the public internet, which is what you want.
3. Pick a location close to your guests, e.g. `asia-south1` (Mumbai) or `eur3` (Europe). **This cannot be changed later.**
4. Click **Create** and wait for provisioning (~30 seconds).

1.2 Generate the service-account key

> OPEN SERVICE ACCOUNTS — DIRECT LINK
> https://console.firebase.google.com/project/bonaca-5689e/settings/serviceaccounts/adminsdk

1. Confirm you are on the **Firebase Admin SDK** tab.
2. Make sure the language toggle reads **Node.js**.
3. Click the blue **Generate new private key** button.
4. A dialog warns you to store the key securely — click **Generate key**.
5. A JSON file downloads, named something like `bonaca-5689e-firebase-adminsdk-a1b2c-3d4e5f6789.json`.

> **This file is a master key to your database**
> It grants full read/write to all of Firestore, bypassing every security rule. Never commit it, never email it, never paste it into a chat or an AI tool. Delete it from your Downloads folder once you have copied the two values out.

1.3 Open the JSON and find the two fields

Open the downloaded file in VS Code. It looks like this — the two fields you need are highlighted:

```
{
  "type": "service_account",
  "project_id": "bonaca-5689e",
  "private_key_id": "a1b2c3d4e5f6...",
  "private_key": "-----BEGIN PRIVATE KEY-----\nMIIEvQ...\n-----END PRIVATE KEY-----\n",   <-- #2
  "client_email": "firebase-adminsdk-a1b2c@bonaca-5689e.iam.gserviceaccount.com",   <-- #1
  "client_id": "1234567890...",
  ...
}
```

1.4 Copy `client_email` → FIREBASE_CLIENT_EMAIL

Copy the value **without** the surrounding quotes. In `.env`:

```
# before
FIREBASE_CLIENT_EMAIL=

# after
FIREBASE_CLIENT_EMAIL=firebase-adminsdk-a1b2c@bonaca-5689e.iam.gserviceaccount.com
```

No quotes, no spaces around the =.

1.5 Copy `private_key` → FIREBASE_PRIVATE_KEY

This is the step people get wrong. Read it carefully.

Select the entire value **including both double quotes** — from the opening " before `-----BEGIN` all the way to the closing " after the final `\n`. Paste it directly after the =:

```
# correct — ONE long line, quotes kept, \n left as literal backslash-n
FIREBASE_PRIVATE_KEY="-----BEGIN PRIVATE KEY-----\nMIIEvQIBADANBgkq...\n-----END PRIVATE KEY-----\n"
```

Three things that will break it

- **Do not** press Enter inside the key. It must stay on one physical line, however long.
- **Do not** convert `\n` into real line breaks. The code at `src/lib/server/firebase.ts:20` does that conversion for you at runtime.
- **Do not** drop the wrapping quotes. Without them the `\n` sequences are mangled.

Your editor will show one very long line that scrolls far to the right. That is correct.

1.6 Restart and verify

Environment variables are read once at server start, so a restart is required — hot reload will not pick them up.

```
# stop the dev server with Ctrl+C, then
npm run dev
```

1. Open `http://localhost:3000/admin` and log in.
2. Find the **“What is on the site right now”** card.
3. The **Content store** row should now read **Firebase Firestore**. If it still says *Local file*, one of the two values is wrong — see Part 5.

Confirmed working

Edit any page in the CRM, hit save, then refresh the Firestore console. You should see a `bonaca-cms` collection containing a `site-content` document.

2 Cloudflare R2 — 5 values

R2 is where guest-facing photography lives. Five values, gathered from three different screens in the Cloudflare dashboard.

2.1 Create the account and enable R2

> SIGN UP (SKIP IF YOU HAVE AN ACCOUNT)
> `https://dash.cloudflare.com/sign-up`

> R2 OVERVIEW
> `https://dash.cloudflare.com/?to=/:account/r2/overview`

1. Sign up and verify your email address.
2. Click **R2 Object Storage** in the left sidebar.
3. Cloudflare asks for a payment method before enabling R2. Add a card.

> **On the card requirement**
> R2's free tier covers 10 GB of storage, 1 million writes and 10 million reads per month — comfortably more than a villa site will use. The card is for overage only. Crucially, R2 charges **zero egress fees**, which is exactly why it suits image hosting.
> Pricing: `https://developers.cloudflare.com/r2/pricing/`

2.2 Create the bucket → R2_BUCKET

1. On the R2 page click **Create bucket**.
2. Name it `bonaca-media`. Lowercase letters, digits and hyphens only.
3. Choose a location hint near your guests — or leave **Automatic**.
4. Leave the storage class as **Standard**.
5. Click **Create bucket**.

```
R2_BUCKET=bonaca-media
```

2.3 Find the Account ID → R2_ACCOUNT_ID

A 32-character hexadecimal string identifying your Cloudflare account.

1. On the R2 overview page, look at the **right-hand sidebar** for **Account ID**.
2. Click the copy icon beside it.
3. Alternatively, read it from your browser's address bar — it is the long hex segment in `dash.cloudflare.com/<account-id>/r2`.

```
R2_ACCOUNT_ID=8f14e45fceea167a5a36dedd4bea2543
```

The code builds the S3 endpoint from this as `https://<account-id>.r2.cloudflarestorage.com` — see `src/lib/server/media.ts:59`. Paste the ID only, never the full URL.

2.4 Create an API token → ACCESS_KEY_ID + SECRET

> **R2 API TOKENS**
> https://dash.cloudflare.com/?to=/:account/r2/api-tokens

1. Click **Create API token** (labelled *Manage API tokens* on some screens).
2. Name it something traceable, e.g. `bonaca-site-uploads`.
3. Under **Permissions** choose **Object Read & Write**. Do not grant Admin — the app only ever puts, lists and deletes objects.
4. Under **Specify bucket**, scope it to **only** `bonaca-media`. A leaked token then cannot touch anything else.
5. Leave TTL as **Forever** unless you plan to rotate on a schedule.
6. Click **Create API Token**.

The result screen shows several values. You need exactly two — the ones under the **S3 Client** heading:

```
R2_ACCESS_KEY_ID=<the "Access Key ID" value>
R2_SECRET_ACCESS_KEY=<the "Secret Access Key" value>
```

> **The secret is displayed exactly once**
> Copy it into `.env` before leaving that page. If you navigate away without copying, the token cannot be recovered — delete it and create a new one.

Ignore the "Token value" and the jurisdiction-specific endpoints on that page; this app does not use them.

2.5 Make the bucket public → R2_PUBLIC_BASE_URL

Uploads must be readable by browsers. By default an R2 bucket is entirely private.

1. Open your `bonaca-media` bucket, then the **Settings** tab.
2. Scroll to **Public access** → **R2.dev subdomain**.
3. Click **Allow Access** and type `allow` to confirm.
4. Cloudflare shows a URL like `https://pub-a1b2c3d4e5f6.r2.dev`. Copy it.

```
# no trailing slash — the code strips one, but keep it clean
R2_PUBLIC_BASE_URL=https://pub-a1b2c3d4e5f6.r2.dev
```

> **Custom domain instead (optional, better for production)**
> In the same Public access panel, **Connect Domain** lets you serve images from `media.bonaca.com`. It needs the domain's DNS on Cloudflare, gives you real CDN caching, and avoids r2.dev's rate limits. If you do this, set `R2_PUBLIC_BASE_URL=https://media.bonaca.com` instead.
> Both `**.r2.dev` and custom hosts are already allowed in `next.config.ts`.

2.6 Restart and verify

```
# Ctrl+C, then
npm run dev
```

1. Open `http://localhost:3000/admin/media`.
2. The badge at the top should read **Cloudflare R2**, not the amber *Local folder*.
3. Upload a test photograph. It should appear, and its URL should begin with your `R2_PUBLIC_BASE_URL`.
4. Confirm the object landed under the `bonaca/` prefix in the Cloudflare bucket browser.

> **All five correct**
>
> The badge only flips to R2 when *all* of account ID, access key, secret, bucket and public URL are present — see `src/lib/server/media.ts:32`. Seeing "Cloudflare R2" proves every one of them parsed.

3 Every link, in one place

Bookmark-ready. The Firebase links are pre-filled with your project ID.

WHAT	LINK
Firestore database	`https://console.firebase.google.com/project/bonaca-5689e/firestore`
Service accounts key download	`https://console.firebase.google.com/project/bonaca-5689e/settings/serviceaccounts/adminsdk`
Project overview	`https://console.firebase.google.com/project/bonaca-5689e/overview`
Project settings	`https://console.firebase.google.com/project/bonaca-5689e/settings/general`
Firestore usage / billing	`https://console.firebase.google.com/project/bonaca-5689e/usage`
Manage service accounts Google Cloud IAM — to revoke keys	`https://console.cloud.google.com/iam-admin/serviceaccounts?project=bonaca-5689e`
Cloudflare R2 overview	`https://dash.cloudflare.com/?to=/:account/r2/overview`
R2 API tokens	`https://dash.cloudflare.com/?to=/:account/r2/api-tokens`
Cloudflare sign-up	`https://dash.cloudflare.com/sign-up`
R2 pricing	`https://developers.cloudflare.com/r2/pricing/`
R2 public buckets docs	`https://developers.cloudflare.com/r2/buckets/public-buckets/`
R2 S3 API token docs	`https://developers.cloudflare.com/r2/api/tokens/`
Firebase Admin SDK docs	`https://firebase.google.com/docs/admin/setup`
Vercel env variables	`https://vercel.com/docs/environment-variables`

The finished `.env`, shape only

```
# CRM login
ADMIN_USERNAME=admin
ADMIN_PASSWORD=<something strong — see Part 4>
ADMIN_SESSION_SECRET=<already set>

# Firebase — Part 1
FIREBASE_PROJECT_ID=bonaca-5689e
FIREBASE_CLIENT_EMAIL=firebase-adminsdk-xxxxx@bonaca-5689e.iam.gserviceaccount.com
FIREBASE_PRIVATE_KEY="-----BEGIN PRIVATE KEY-----\n...\n-----END PRIVATE KEY-----\n"
FIREBASE_CONTENT_COLLECTION=bonaca-cms
FIREBASE_CONTENT_DOCUMENT=site-content

# Cloudflare R2 — Part 2
R2_ACCOUNT_ID=<32 hex chars>
R2_ACCESS_KEY_ID=<from API token>
R2_SECRET_ACCESS_KEY=<from API token>
R2_BUCKET=bonaca-media
R2_PUBLIC_BASE_URL=https://pub-xxxxxxxx.r2.dev
R2_PREFIX=bonaca

# Public site
NEXT_PUBLIC_SITE_URL=http://localhost:3000
```

4 Before you go live

Four things that are not credentials to fetch, but will bite you if skipped.

The admin password is `12345678`

That is the only thing standing between the public internet and full edit access to the site. Change it now:

```
# generate something strong
openssl rand -base64 24
```

Put the result in `ADMIN_PASSWORD` and store it in a password manager.

Copy all 15 values into Vercel

`.env` is git-ignored — I verified this, and the file is not tracked. That is correct, but it means your host has no idea these values exist. Before the first production deploy, add every one of them under **Project** → **Settings** → **Environment Variables**.

> **The private key needs care in the Vercel UI**
>
> Paste it exactly as it appears in `.env` — wrapping quotes included, on one line, `\n` intact. Vercel's textarea will accept a real newline silently and the deploy will then fail with an opaque *"Failed to parse private key"*.

Set the real site URL at deploy time

`NEXT_PUBLIC_SITE_URL` feeds canonical tags, Open Graph URLs and the sitemap. Left at `localhost:3000` in production it will actively damage your search ranking. In Vercel set it to `https://bonaca.com` (or whichever domain is live) for the Production environment only — keep localhost for local development.

Delete the downloaded JSON

Once Part 1 is verified, remove the service-account file from `~/Downloads`. If it ever leaks, revoke it from the Google Cloud IAM link in Part 3 — the key can be deleted there and a fresh one generated, and the old one stops working immediately.

One cosmetic warning you will see in the terminal

Unrelated to any of the above, and harmless:

```
Image with src "/images/bonaca/hero/hero-wide.jpg" is using quality "88"
which is not configured in images.qualities [75]
```

Next.js 16 requires every `quality` value to be declared up front. Add 88 to `images.qualities` in `next.config.ts` to silence it.

5 If something does not work

SYMPTOM	CAUSE AND FIX
Admin still shows **"Local file"**	All three of project ID, client email and private key must be non-empty. Most often the private key still holds the placeholder `""`, or the server was not restarted.
`Failed to parse private key`	The `\n` sequences became real line breaks, or the wrapping quotes were dropped. Re-copy the value from the JSON exactly as described in step 1.5.
`16 UNAUTHENTICATED` or `PERMISSION_DENIED`	The key was revoked, or belongs to a different Firebase project than `FIREBASE_PROJECT_ID`. Generate a fresh key from the same project.
`5 NOT_FOUND` on save	The Firestore database was never created. Do step 1.1.
Media page still shows **"Local folder"**	All five R2 values are required together; any one blank falls back to local. Check for a stray trailing space, and that `R2_PUBLIC_BASE_URL` includes `https://`.
Uploads succeed but images show broken	The bucket is not public. Do step 2.5 — without the r2.dev subdomain enabled, objects return 401 to browsers even though the upload itself worked.
`SignatureDoesNotMatch`	Secret access key mistyped or truncated. It cannot be re-read — delete the token and create a new one.
`NoSuchBucket`	`R2_BUCKET` does not match the real bucket name, or `R2_ACCOUNT_ID` is wrong so the endpoint points at the wrong account.
Changes to `.env` have no effect	Next.js reads environment variables at process start. Hot reload does not pick them up — stop the server with Ctrl+C and run `npm run dev` again.

Quick sanity check from the terminal

Confirms which keys are still empty without printing any secret:

```
grep -E '^[A-Z_]+=' .env | awk -F= '{ if ($2 == "" || $2 == "\"\"") print "EMPTY: " $1 }'
```

> **Verified against the code, not from memory**
> Every variable name, fallback behaviour, file path and line reference in this guide was read out of your repository on 7 September 2026. The status pills in Part 0 reflect the actual contents of your `.env` at that moment.